

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2020

Subject: 392133 Physics III

Section: 15-18

Date: 2 September 2020

Time: 10.00-12.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Directions: The test is designed to measure your comprehension. The test is divided into 6 sections. There will be 7 pages (including this page) and they are worth 60 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

- 1.1 A block-spring system consists of a spring with constant  $k = 50 \text{ N/m}$  attached to a 500-g block on a frictionless surface. The block is pulled 2.60 cm from equilibrium and released from rest. For the resulting oscillation, find the (a) angular frequency, (b) period, and (c) velocity at the displacement  $x = 1 \text{ cm}$ . (6 points) [All values are required in SI unit]

- 1.2 A 0.5-kg block resting on a frictionless, horizontal surface is attached to a spring whose constant is 0.50 N/m. A horizontal force  $F$  causes the spring to stretch a distance of 10 cm from its equilibrium position. Find the total energy stored in the system when the spring is stretched (2 points), and the kinetic energy when its displacement is a half of amplitude. (2 points)

- 2.1 A vertical spring stretches 4.9 cm when a 10-g object is hung from it. The object is replaced with a block of mass 20 g that oscillates up and down in simple harmonic motion. Calculate the angular frequency of motion. (4 points)  
Given the gravitational acceleration is  $9.80 \text{ m/s}^2$ .

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- 2.2 The position of a particle is given by the expression  $x = 4.00 \cos(3\pi t + \pi)$ , where  $x$  is in meter and  $t$  is in second. Determine (a) period of the motion, (b) the position of the particle at  $t = \frac{5}{9}$  s. and (c) the velocity of the particle at a position as you calculated in (b). (6 points)

3.1 The period of a simple pendulum is 1 s on Earth. When brought to a planet where  $g$  is one-ninth that on Earth, its period becomes: (5 points)

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3.2 When a particular wire is vibrating with a frequency of 4.00 Hz, a transverse wave of wavelength 60.0 cm is produced. Determine the speed of waves along the wire. (5 points)

4. A string is stretched and fixed at both ends, 200 cm apart. If the density of the string is 0.015 g/cm, and its tension is 600 N,
- (a) what is the wavelength (in cm) of the first harmonic? (3 points)
  - (b) what is the fundamental frequency? (5 points)
  - (c) what is the frequency of the second harmonic? (2 points)

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5.1 The area of a typical eardrum is about  $5.00 \times 10^{-5} \text{ m}^2$ . (a) Calculate the average sound power incident on an eardrum at the threshold of pain, which corresponds to an intensity of  $1.00 \text{ W/m}^2$ . (b) How much energy is transferred to the eardrum exposed to this sound for 1.00 min?

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5.2 A speaker produces the sound intensity level of 60 dB as measured by a listener. If the power of this speaker is increased by a factor of 10, what will be the sound intensity level measured by the listener?



6.1 A railroad train is traveling at 20.0 m/s in still air. The frequency of the note emitted by the locomotive whistle is 400 Hz. What is the frequency of the sound heard by a stationary listener (a) in front of the locomotive and (b) behind the locomotive?  
Given the speed of sound in air is 340 m/s.

6.2 The fundamental frequency of a pipe that is open at both ends is 680 Hz. (a) How long is this pipe? If one end is now closed, find (b) the wavelength and (c) the frequency of the new fundamental.  
Given the speed of sound in air is 340 m/s.

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